Ugochukwu Okonkwo

Melbourne, FL (Open to Relocate) | 267-968-3159 | ugochukwuokonkwo04@gmail.com |
linkedin.com/in/ugochukwu-okonkwo-7859522a1/ | ugochukwuokonkwo.github.io/

PROFESSIONAL SUMMARY

Biomedical Engineer with hands-on experience in medical device design, CAD modeling, prototyping, usability testing, and regulated product development. Experienced in supporting orthopedic implant and surgical instrument development, DV&V activities, engineering documentation, and cross-functional collaboration within MedTech environments. Combines technical engineering expertise with business and project management knowledge to support the development of innovative, patient-centered medical devices.

RELEVANT WORK EXPERIENCE

Globus Medical, Audubon, PA

Project Engineer Co-op, Jan 2024 - Aug 2024

- Applied Creo Parametric and engineering drawing standards to support development of orthopedic implants and surgical instruments in a regulated MedTech environment
- Utilized rapid prototyping, DFM principles, and cross-functional collaboration to support design iterations, manufacturability improvements, and functional evaluations
- Supported DV&V activities, engineering change processes, and technical documentation aligned with FDA and MDR quality system requirements
- Developed Excel macro-based inspection and traceability tools to improve quality documentation efficiency and workflow standardization
- Coordinated with manufacturing, operations, prototype shops, and suppliers to support prototype builds, testing, and production readiness
- Applied product lifecycle management concepts including DCO processing, UDI tracking, and DHF/DMR documentation support

EDUCATION

Florida Institute of Technology, Melbourne, FL
Ph.D. in Biomedical Engineering, Expected May 2030
Widener University, Chester, PA
Master of Science in Engineering in Biomedical Engineering (Thesis)**,** Aug 2026
Thesis: Ergonomic Redesign of an Assistive Button Hook for Individuals with Dexterity Impairments
Master of Business Administration in Management, May 2026
Bachelor of Science in Biomedical Engineering, May 2025
Minors: Management, Mathematics, Mechanical & Robotics Engineering

CONFERENCE PRESENTATIONS

Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL | October 2026
Human-Centered Redesign of a Button Hook for Individuals with Dexterity Impairments: Development and Preliminary Usability Evaluation
Poster Presentation.

RESEARCH EXPERIENCE

Master's Thesis Research

Widener University | Aug 2025 – Jul 2026

- Designed and developed an ergonomic assistive medical device from concept through prototype evaluation, applying human factors engineering and user-centered design principles to improve usability for individuals with hand dexterity impairments.
- Generated and evaluated multiple design concepts using CAD, rapid prototyping, and iterative engineering reviews to optimize ergonomics, manufacturability, and functional performance.
- Developed custom manufacturing fixtures, wire-forming molds, and assembly processes using FDM 3D printing, CO_2 laser cutting, and precision fabrication techniques to improve prototype consistency.
- Conducted design verification through expert evaluations and structured usability testing, utilizing QUEST 2.0-based surveys and user feedback to assess comfort, ease of use, grip stability, and mechanism performance.
- Collaborated with interdisciplinary clinical stakeholders, including Physical Therapy and Occupational Therapy professionals, to translate user needs into measurable engineering design requirements.
- Analyzed quantitative and qualitative usability data to validate design decisions, guide iterative improvements, and support recommendations for future assistive device development.

Low-Cost Versatile Electrospinning System
Undergraduate Senior Research Project, Widener University | Aug 2024 – April 2025
- Applied CAD modeling, prototype fabrication, and mechanical assembly techniques to develop a low-cost electrospinning platform for nanofiber production
- Utilized bandsaws, table saws, and fabrication tools to machine acrylic panels and aluminum structural components for custom system integration
- Performed aluminum tapping, fastening, and enclosure assembly operations to support structural stability and modular component integration
- Incorporated iterative prototyping, 3D printing, and system integration methods to refine syringe pump and rotating collector assemblies
- Conducted experimental parameter optimization involving flow rate, collector speed, and polymer delivery to improve fiber alignment and consistency
- Implemented safety-focused engineering controls including insulated acrylic housing, grounding, and controlled high-voltage operation

ADDITIONAL WORK EXPERIENCE
Widener University Office of Student Success, Chester, PA
Graduate Student Employee, Aug 2025 - Jun 2026
- Coordinated tutor-student matching and supported front desk operations, improving service efficiency, scheduling, and student support delivery.
- Trained and onboarded new tutors, reinforcing academic support protocols and effective communication strategies.
- Analyzed tutoring and engagement data using Microsoft Excel (pivot tables, queries) and developed promotional materials to increase program effectiveness and student participation.

Peer & Math Tutor, Sep 2024 - May 2025
- Delivered one-on-one and group tutoring across courses (MGT-210, MATH-242, ENGR-213), adapting teaching strategies to diverse learning styles.
- Simplified complex mathematical and technical concepts while guiding students through problem-solving, projects, and exam preparation.
- Fostered student confidence, independent learning, and problem-solving skills.

Widener University Office of Admissions, Chester, PA
Crew Leader, Aug 2023 - Dec 2023
- Led orientation programs and delivered presentations to groups of 20+ students, supporting their transition into university life.
- Collaborated with admissions and student affairs teams during university events and outreach initiatives

CERTIFICATIONS
Certified SOLIDWORKS Professional (CSWP)
Certified SOLIDWORKS Sheet Metal Professional (CSWPA-SM)
PTC University Creo User Modeling Specialist Certification
PTC University Creo User Documentation Specialist
ISO 14971: Medical Device Risk Management
Medical Device Process Validation (ISO 13485, IQ/OQ/PQ)
Lean Six Sigma Green Belt

TECHNICAL SKILLS
- CAD & Design: Creo Parametric, Autodesk Inventor, SolidWorks, 3D Printing, Iterative Design, Prototyping
- Modeling & Simulation: ANSYS Fluent, Thermal & Bioheat Modeling, Mesh Independence Studies
- Data & Analysis: MATLAB, Excel (Regression, Pivot Tables, Data Analysis ToolPak), SQL, Tableau
- Engineering & Medical Device: Design Verification & Validation (V&V), Human-Centered Design, Ergonomics, Usability Testing, Quality Systems (QMS), CO_2 Laser Cutting, Bandsaw & Table Saw Operation
- Quality & Regulatory: Lean Six Sigma, DMAIC, FMEA, Process Improvement, Design Controls
- Laboratory Techniques: Bradford Assay, Protein Extraction (RIPA Buffer)

PROFESSIONAL SKILLS
Research communication • Technical writing • Project management • Mentorship • Leadership • Collaboration & Teamwork • Problem-solving & Critical thinking